1 Basics

- Taylor: $f(a)+\frac{f'(a)}{1!}(x-a)+\frac{f''(a)}{2!}(x-a)^2+\dots$
  - $f(\boldsymbol{a})+\frac{\partial f(\boldsymbol{x})}{\partial \boldsymbol{x}}\Big|_{\boldsymbol{a}}-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{a})^\top\left(\frac{\partial^2 f(\boldsymbol{x})}{\partial \boldsymbol{x}\partial \boldsymbol{x}^\top}\right)\Big|_{\boldsymbol{a}}(\boldsymbol{x}-\boldsymbol{a})$
  - Power series of exp.: $\exp(x):=\sum_{k=0}^{\infty}\frac{x^k}{k!}$
- Entropy: $H(X)=\mathbb{E}_X[-\log\mathbb{P}(X=x)]$
- Diverg.: $D_{\mathrm{KL}}(P\,\|\,Q)=\sum_{x\in\mathcal{X}}P(x)\log\left(\frac{P(x)}{Q(x)}\right)\ge 0$
- $1-z\le\exp(-z)$
- Cauchy-Schwarz: $|\mathbb{E}[X,Y]|^2\le\mathbb{E}[X^2]\mathbb{E}[Y^2]$
- Jensen, $f(X)$ convex: $f(\mathbb{E}[X])\le\mathbb{E}[f(X)]$

1.1 Probability / Statistics

- Gaussian: $\mathcal{N}(x\mid\mu,\sigma^2)=\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}\right)$
  $\mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu},\Sigma)=\frac{1}{\sqrt{(2\pi)^d|\Sigma|}}\exp\left(-\frac{1}{2}(\boldsymbol{x}-\mu)^\top\Sigma^{-1}(\boldsymbol{x}-\mu)\right)$
  $X\sim\mathcal{N}(\boldsymbol{\mu},\Sigma),\ Y=A+BX\Rightarrow Y\sim\mathcal{N}(A+B\boldsymbol{\mu},B\Sigma B^\top)$
- Binom.: $f(k,n;p)=\mathbb{P}(X=k)=\binom{n}{k}p^k(1-p)^{n-k}$
- $\mathbb{V}[X]=\mathbb{E}[(X-\mathbb{E}[X])^2]=\mathbb{E}[X^2]-\mathbb{E}[X]^2$
  $\mathbb{V}[X+Y]=\mathbb{V}[X]+\mathbb{V}[Y]+2\mathbb{C}\mathrm{ov}(X,Y)$
- $\mathbb{C}\mathrm{ov}(X,Y)=\mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])]$
  $=\mathbb{E}[XY]-\mathbb{E}[X]\mathbb{E}[Y]$
  $\mathbb{C}\mathrm{ov}(aX,bY)=ab\,\mathbb{C}\mathrm{ov}(X,Y)$

1.2 Calculus

- $\int uv'\,\mathrm{d}x=uv-\int u'v\,\mathrm{d}x$ • $\frac{\partial}{\partial x}\frac{g}{h}=\frac{g'h}{h^2}-\frac{gh'}{h^2}$
- $\frac{\partial}{\partial \boldsymbol{x}}(\boldsymbol{b}^\top A\boldsymbol{x})=A^\top\boldsymbol{b}$ • $\frac{\partial}{\partial \boldsymbol{x}}(\boldsymbol{b}^\top\boldsymbol{x})=\frac{\partial}{\partial \boldsymbol{x}}(\boldsymbol{x}^\top\boldsymbol{b})=\boldsymbol{b}$
- $\frac{\partial}{\partial X}(\boldsymbol{c}^\top X^\top\boldsymbol{b})=\boldsymbol{b}\boldsymbol{c}^\top$ • $\frac{\partial}{\partial X}(\boldsymbol{c}^\top X\boldsymbol{b})=\boldsymbol{c}\boldsymbol{b}^\top$
- $\frac{\partial}{\partial \boldsymbol{x}}(\boldsymbol{x}^\top A\boldsymbol{x})=(A^\top+A)\boldsymbol{x}\overset{A\text{ sym.}}{=}2A\boldsymbol{x}$
- $\frac{\partial}{\partial X}Tr(X^\top A)=A$ • Tr. trick: $\boldsymbol{x}^\top A\boldsymbol{x}\overset{\text{inner prod.}}{=}$
  $Tr(\boldsymbol{x}^\top A\boldsymbol{x})\overset{\text{cyclic permut.}}{=}Tr(\boldsymbol{x}\boldsymbol{x}^\top A)=Tr(A\boldsymbol{x}\boldsymbol{x}^\top)$
- $|X^{-1}|=|X|^{-1}$ • $\frac{\partial}{\partial X}\log|X|=X^{-\top}$ • $\frac{\mathrm{d}}{\mathrm{d}x}|x|=\frac{x}{|x|}$
- $\frac{\partial}{\partial \boldsymbol{x}}\|\boldsymbol{x}\|_2=\frac{\partial}{\partial \boldsymbol{x}}(\boldsymbol{x}^\top\boldsymbol{x})=2\boldsymbol{x}$ • $\frac{\partial}{\partial \boldsymbol{x}}\|\boldsymbol{x}-\boldsymbol{b}\|_2=\frac{\boldsymbol{x}-\boldsymbol{b}}{\|\boldsymbol{x}-\boldsymbol{b}\|_2}$
- $\frac{\partial}{\partial \boldsymbol{x}}\|\boldsymbol{x}\|_1=\mathrm{sgn}(\boldsymbol{x})$ $\mathrm{sgn}(\boldsymbol{x})\in\{\pm1\}^p$ is row-wise
- $\sigma(x)=\frac{1}{1+\exp(-x)}\implies\nabla\sigma(x)=\sigma(x)(1-\sigma(x))$
- $\tanh x=\frac{2\sinh x}{2\cosh x}=\frac{e^x-e^{-x}}{e^x+e^{-x}}$ • $\nabla\tanh x=1-\tanh^2 x$

2 Density Estimation

Bayesianism: Prior $P(\theta)$, likelihood $P(X\mid\theta)$, posterior $P(\theta|x_{1\dots n})$. **Bayes:** $P(\theta|X)=\frac{P(X|\theta)P(\theta)}{P(X)}$, $P(X)=\sum_\theta P(X|\theta_i)P(\theta_i)$

Frequentism: Param. model $P(Y\mid X,\theta)$, likelihood $P(X,Y\mid\theta)$. $\hat\theta_{\mathrm{MLE}}=\arg\max_\theta P(X,Y\mid\theta)$.

Ex. Gaussian: $X_{i=1}^n\sim\mathcal{N}(\mu,\sigma^2)\Rightarrow\log\mathcal{L}_n(\mu,\sigma)=-n\log\sigma-\frac{nS^2}{2\sigma^2}-\frac{n(\bar X-\mu)^2}{2\sigma^2}\Rightarrow\hat\mu=\bar X,\hat\sigma=S$

2.1 Estimation - MLE Properties

Consistency: $\forall\epsilon>0,\ \mathbb{P}\{|\hat\theta_n-\theta^*|>\epsilon\}\overset{n\to\infty}{\longrightarrow}0$

Equivariance: If $\hat\theta_n$ is MLE of θ, then $g(\hat\theta_n)$ is MLE of $g(\theta)$.

Asympt. normality: $\frac{1}{\sqrt{n}}(\hat\theta_n-\theta^*)\to\mathcal{N}(0,J^{-1}(\theta^*)I_n(\theta^*)J^{-1}(\theta^*))$

Asympt. efficiency: $\hat\theta_n$ minimises $\mathbb{E}[(\hat\theta_n-\theta^*)^2]$ as $n\to\infty$, i.e. $\mathbb{E}[(\hat\theta_n-\theta^*)^2]\overset{n\to\infty}{=}\frac{1}{I_n(\theta^*)}$ (Rao Cr.)

Among all consistent estimators $\hat\theta_n$ has *smallest variance*: $\lim_{n\to\infty}(\mathbb{V}[\hat\theta_n]I_n(\theta^*))^{-1}=1$

2.2 Rao Cramer inequality — all $\mathbb{E}$ w.r.t. $P(x\mid\theta^*)$

Score func.: $\boldsymbol{\Lambda}=\frac{\partial\log\mathbb{P}(x|\theta)}{\partial\theta}$, $\mathbb{E}[\boldsymbol{\Lambda}]=0$

Fisher info.: $I_n(\theta)=\mathbb{V}[\boldsymbol{\Lambda}]$

$J(\theta)=\mathbb{E}[\boldsymbol{\Lambda}^2]=-\mathbb{E}\left[\frac{\partial^2\log\mathbb{P}(x|\theta)}{\partial\theta\partial\theta^\top}\right]=-\mathbb{E}\left[\frac{\partial\boldsymbol{\Lambda}}{\partial\theta}\right]$

General bound: $\mathbb{E}[(\hat\theta_n-\theta^*)^2]\ge\frac{\left(1+\frac{\partial}{\partial\theta}b_{\hat\theta}\right)^2}{\mathbb{E}[\boldsymbol{\Lambda}^2]}+b_{\hat\theta}^2$

Unbiased case: $\mathbb{E}[(\hat\theta_n-\theta^*)^2]=\mathbb{V}[\hat\theta_n]\ge\frac{1}{I_n(\theta^*)}$

Tradeoff: $\mathbb{E}[(\hat\theta_n-\theta^*)^2]=\mathbb{V}[\hat\theta_n]+\mathrm{bias}^2(\hat\theta_n)$

Bias: $\mathrm{bias}(\hat\theta_n)\equiv b_{\hat\theta}(\theta^*)=\mathbb{E}[\hat\theta_n]-\theta^*\overset{\text{unbiased}}{=}0$

3 (Linear) Regression — model: $\hat{y}=X\beta$

Assuming $\boldsymbol{X}^\top\boldsymbol{X}$ non-singular.

Bayesian view: $(Y\mid X,\beta)\sim\mathcal{N}(\boldsymbol{x}^\top\beta,\sigma^2\mathbb{I})$

Distrib. of estimator $\hat\beta_{\mathrm{LS}}\sim\mathcal{N}(\beta,(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\sigma^2)$

Ridge: $\epsilon_{\mathrm{RSS}}(\beta,\lambda)=(\boldsymbol{y}-\boldsymbol{X}^\top\beta)^\top(\boldsymbol{y}-\boldsymbol{X}^\top\beta)+\lambda\beta^\top\beta$

$\hat\beta=(\boldsymbol{X}^\top\boldsymbol{X}+\lambda\mathbb{I})^{-1}\boldsymbol{X}^\top\boldsymbol{y}$, prior: $\beta\sim\mathcal{N}(0,\frac{\sigma^2}{\lambda}\mathbb{I})$

(Ridge) Shrinkage: Decompose $\boldsymbol{X}=\boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^\top$

$\boldsymbol{X}\hat\beta=\boldsymbol{U}\boldsymbol{D}(\boldsymbol{D}^2+\lambda\mathbb{I})^{-1}\boldsymbol{D}\boldsymbol{U}^\top\boldsymbol{y}=\sum_{j\le d}\boldsymbol{u}_j\frac{d_j^2}{d_j^2+\lambda}\boldsymbol{u}_j^\top\boldsymbol{y}$

Lasso: $\hat\beta=\arg\min_\beta\sum_{i\le n}(y_i-x_i^\top\beta)^2+\lambda\|\beta\|_1$

(no closed form), prior: $p(\beta_i)=\frac{\lambda}{4\sigma^2}\exp(-|\beta_i|\frac{\lambda}{2\sigma^2})$

Bias-variance: $\mathrm{MSE}=\mathbb{E}_{D,X,Y}[(\hat f(x)-Y)^2]$

$=(\mathbb{E}_{D,X}[\hat f(x)]-\mathbb{E}_Y[])]^2+\mathbb{E}_D[(\mathbb{E}_{D,X}[\hat f(x)]-\hat f(x))^2]+\mathbb{E}[(\mathbb{E}_Y[-]Y)^2]$

$=\mathrm{bias}^2+\mathrm{variance}+\mathrm{noise}$

Gauss-Markov Theorem:

For any linear estimator $\widetilde{\theta}=c^\top\boldsymbol{y}=a^\top(\hat\beta+\boldsymbol{D}\boldsymbol{y})$ that is unbiased for $a^\top\beta$, it holds: $\mathbb{V}[a^\top\hat\beta]\le\mathbb{V}[c^\top\boldsymbol{y}]$.

Among all linear **u**-estimators, $\hat\beta_{\mathrm{LS}}$ minimises the gen. error!

3.1 3.1 Causality

Shortcut Learning: Model uses spurious correlation $X\leftrightarrow Y$ (e.g. grass ↔ cow)

$X_\alpha^\perp$: Part of X not causally influenced by α

Causal: $X_W^\perp$ and $X,W\&Y$ are causes of Y.

Anti-Causal: $X_W^\perp$ and $X,W\&Y$ are caused by Y.

W non-causal feature (e.g. background) influences X but not Y. **Counterfactually Inv.:** $f(X(w))=f(X(w'))$ for $w,w'\in W$. **Confounding:** Hidden U affects W,X. **Selection:** Hidden S filters on W,X. **Theorem For Counterfactual Invariance (CI):** Anti-causal: $f(X)\perp W|Y$ Causal (no selection): $f(X)\perp W$ Causal (no confounding): $Y\perp X|X^{\perp W},W$ and $f(X)\perp W|Y$.

4 Uncertainty Quantification

4.1 Bayesian Linear Regression

Model: $\boldsymbol{y}=\boldsymbol{X}^\top\beta+\epsilon$, with $\epsilon\sim\mathcal{N}(\epsilon\mid0,\sigma^2\mathbb{I})$

Likelihood: $P(\boldsymbol{y}\mid\boldsymbol{X},\beta,\sigma)=\mathcal{N}(\boldsymbol{y}\mid\boldsymbol{X}^\top\beta,\sigma^2\mathbb{I})$

Prior: $P(\beta\mid\boldsymbol{\Lambda})=\mathcal{N}_d(\beta\mid0,\boldsymbol{\Lambda}^{-1})$

(Ridge regr. if $\boldsymbol{\Lambda}=\lambda\mathbb{I}$ and $\sigma=1$)

Posterior: $P(\beta\mid\boldsymbol{X},\boldsymbol{y},\boldsymbol{\Lambda})=\mathcal{N}(\beta\mid\mu_\beta,\Sigma_\beta)$

with $\mu_\beta=(\boldsymbol{X}^\top\boldsymbol{X}+\sigma^2\boldsymbol{\Lambda})^{-1}\boldsymbol{X}^\top\boldsymbol{y}$

and $\Sigma_\beta=\sigma^2(\boldsymbol{X}^\top\boldsymbol{X}+\sigma^2\boldsymbol{\Lambda})^{-1}$

$\implies\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{y}\mid0,\ \boldsymbol{X}\boldsymbol{\Lambda}^{-1}\boldsymbol{X}^\top+\sigma^2\mathbb{I})$ using $\mathbb{E}_{\beta,\epsilon}[\,\cdot\,]$

kernel $k(x_i,x_j):=x_i^\top\boldsymbol{\Lambda}^{-1}x_j$

4.2 Gaussian Process

$\boldsymbol{y}\sim\mathcal{N}(\boldsymbol{y}\mid m(\boldsymbol{X}),K(\boldsymbol{X},\boldsymbol{X})+\sigma^2\mathbb{I})$

$\begin{bmatrix}\boldsymbol{y}\\y_{n+1}\end{bmatrix}\sim\mathcal{N}\left(\begin{bmatrix}\boldsymbol{y}\\y_{n+1}\end{bmatrix}\middle|\begin{bmatrix}m(\boldsymbol{X})\\m(x_{n+1})\end{bmatrix},\begin{bmatrix}C_n&\boldsymbol{k}\\\boldsymbol{k}^\top&c\end{bmatrix}\right)$

$p(y_{n+1}\mid x_{n+1},\boldsymbol{X},\boldsymbol{y})=\mathcal{N}(y_{n+1}\mid\mu_{n+1},\sigma^2_{n+1})$

with $\mu_{n+1}=m(x_{n+1})+\boldsymbol{k}^\top C_n^{-1}(\boldsymbol{y}-m(\boldsymbol{X}))$

and $\sigma^2_{n+1}=c-\boldsymbol{k}^\top C_n^{-1}\boldsymbol{k}$

where $\boldsymbol{K}=k(\boldsymbol{X},\boldsymbol{X})$, $\boldsymbol{k}=k(x_{n+1},\boldsymbol{X})$,

$C_n=\boldsymbol{K}+\sigma^2\mathbb{I}$, $c=k(x_{n+1},x_{n+1})+\sigma^2$

4.3 Kernels — scalar product $K_{ij}=k(\boldsymbol{x}_i,\boldsymbol{x}_j)$

Valid kernel: must be **symmetric** and **p.s.d.** ($\boldsymbol{x}^\top\boldsymbol{K}\boldsymbol{x}\ge0\ \forall\boldsymbol{x}$ *or* pos. eigenvalues *or* pos. principal minors). Must have a (pot. ∞-dim.) **feature vector** ϕ s.t. $k(\boldsymbol{x},\boldsymbol{x}')=\phi(\boldsymbol{x})^\top\phi(\boldsymbol{x}')$.

Common kernels:

Linear: $\boldsymbol{x}^\top\boldsymbol{x}'$

Polynomial: $(\boldsymbol{x}^\top\boldsymbol{x}'+1)^p$, $p\in\mathbb{N}$

RBF (Gaussian): $\exp(-\|\boldsymbol{x}-\boldsymbol{x}'\|_2^2/h^2)$

Sigmoid: $\tanh(\kappa\cdot\boldsymbol{x}^\top\boldsymbol{x}'-b)$

Kernel construction: • k_1+k_2 • $c\cdot k_1$, $c>0$

- $k_1\cdot k_2$ • $f(\boldsymbol{x})k_1(\boldsymbol{x},\boldsymbol{x}')f(\boldsymbol{x}')$
- $k(\phi(\boldsymbol{x}),\phi(\boldsymbol{x}'))$ with $\phi:\mathcal{X}\to\mathbb{R}^d$
- $g(k_1)$ with g: exp. *or* polyn. w/ all pos. coeff.

4.4 Active Learning — $I(X;Y)=H(X)-H(X|Y)$

Mutual Info: $I(X;Y)=\sum_{x,y}p(x,y)\log\frac{p(x,y)}{p(x)p(y)}$

Transd.-L.: $x_n=\arg\max_{x\in S}I(f_{x\in A};y_x|D_{n-1})$

→ For $f\sim\mathrm{GP}(\mu,k)$: $I=\frac{1}{2}\log\frac{\mathrm{Var}(y_x|D_{n-1})}{\mathrm{Var}(y_x|f_{x\in A},D_{n-1})}$

Safe BO: GP s_{f^*},g^* observe $y_i=f^*(x_i),z_i=g^*(x_i)$

95% conf.: $[l_n^f,u_n^f],[l_n^g,u_n^g]$

$S_n=\{x:l_n^g(x)\ge0\},\hat S_n=\{x:u_n^g(x)\ge0\}$

$A_n=\{x\in\hat S_n:u_n^f(x)\ge\max_{x'\in S_n}l_n^f(x')\}$

Now ITL on sample space S_n, target space A_n.

Batch Active Learning: $B_\delta(x)=\{x':\|x-x'\|\le\delta\}$

$\Rightarrow\arg\max_{L\subseteq X,|L|=b}\Pr(\cup_{x\in L}B_\delta(x))$

$\Rightarrow\arg\max_{L\subseteq X,|L|=b}\frac{1}{|X|}|\{x':\|x'-x\|\le\delta,x\in L\}|$

NP-hard Problem, greedy algo (ProbCover):

$L=\emptyset,E=\{(x,x'):\|x-x'\|\le\delta\}$ For $i=1,\dots b$:

Add $x_i=\arg\max_{x\in X}|\{x':(x,x')\in E,x'\in X\}|$

Update $L\leftarrow L\cup\{x_i\},E\leftarrow E\setminus(\{x_i\}\times(B_\delta(x_i)\cap X))$

5 Support Vector Machine (SVM)

Primal (soft margin): $\min_{\boldsymbol{w},w_0,\xi}\frac{1}{2}\|\boldsymbol{w}\|^2+C\sum_i\xi_i$

s.t. $y_i(\boldsymbol{w}^\top\boldsymbol{x}_i+w_0)\ge1-\xi_i$ and $\xi_i\ge0$

↪ intractable if $\varphi(\boldsymbol{x}_i)$ instead of $\boldsymbol{x}_i$

↪ $\xi_i=0$ means $\boldsymbol{x}_i$ was not neglected

Dual: $\max_{\boldsymbol{\alpha}}\sum_i\alpha_i-\frac{1}{2}\sum_{(i,j)}\alpha_i\alpha_jy_iy_j\boldsymbol{x}_i^\top\boldsymbol{x}_j$

s.t. $0\le\alpha_i\le C$; $\sum_i\alpha_iy_i=0$

↪ solve α via quadratic optimisation

Optimal hyperplane: $\boldsymbol{w}^*=\sum_i\alpha_i^*y_i\boldsymbol{x}_i$

↪ $\alpha_i^*\ne0$ only for *support vectors*

Optimal slack: $\xi_i^*=\max(0,1-y_i(\boldsymbol{w}^{*\top}\boldsymbol{x}_i+w_0^*))$

5.1 Structural SVMs

$\min_{\boldsymbol{w},\xi}\frac{1}{2}\|\boldsymbol{w}\|^2+\frac{C}{n}\sum_{i\le n}\xi_i$ s.t. $\xi_i\ge0$ and $\forall y'\ne y_i$:

$\boldsymbol{w}^\top\Psi(x_i,y_i)\ge\Delta(y_i,y')+\underbrace{\boldsymbol{w}^\top\Psi(x_i,y')}_{\hookrightarrow\text{mislabelings}}-\xi_i-\epsilon$

Ψ: *joint-feature map*; Δ: loss / class dissimilarity func.; $\boldsymbol{w}^\top\Psi(x,y)$: compatibility score btw. x and y; ϵ: tolerance / universal slack variable.

Prediction: $c(x)=\arg\max_y\boldsymbol{w}^\top\Psi(x,y)$

Note: For optimal $\boldsymbol{w}^*,\xi^*$, emp. risk$(\boldsymbol{w}^*)\le\frac{1}{n}\sum_i\xi_i^*$

Training: Start without any constraints.
In each iteration, add for each (x_i,y_i) the constraint with $y'\ne y_i$ that is the "most violated" and solve again with quadr. optimisation.

6 Ensemble Methods — $\hat f(x)\triangleq\frac{1}{B}\sum_{i\le B}\hat f_i(x)$

$\mathrm{bias}[\hat f(x)]=\frac{1}{B}\sum\mathrm{bias}[\hat f_i(x)]$

$\mathbb{V}[\hat f]=\frac{1}{B^2}\sum\mathbb{V}_D[\hat f_i]+\frac{1}{B^2}\sum\sum_{i,j}\mathbb{C}\mathrm{ov}((\hat f_i,\hat f_j)\approx\frac{\sigma^2}{B}$

6.1 Bagging (Bootstrap aggregation)

1. Draw M bootstrap sets $Z_1^*,\dots,Z_M^*$
2. Train M base models $b^{(1)},\dots,b^{(M)}$
3. **Aggregate:** $\mathfrak{b}^{(M)}(\boldsymbol{x})=\begin{cases}\frac{1}{M}\sum_{t\le M}b^{(t)}(\boldsymbol{x})&\text{regr.}\\\mathrm{sign}(\sum_t b^{(t)}(\boldsymbol{x}))&\text{class.}\end{cases}$

Why it works: Small *variance* (weak learners)
Random Forest: At each splitting step, u.a.r. choose m of p features & split only one (best) feature. → reduce *correlation* between trees.

6.2 Boosting

Sequentially train weak learners on all data, but ↗ weight of misclass. samples (↘ bias).
AdaBoost: Stat. learning (*forward stagewise additive modeling*) with **exp. loss**, trains max-margin ($= y_i \hat{b}(x_i)$), self-avg. and interpolating (↘ overfitting) classifiers.

[Init]: $w_i \leftarrow 1/n\ \forall i \le n$ for $b = 1 \dots B$:
[Train]: Fit classifier $c_b(x)$ using weights w_i
[Eval]: $\epsilon_b = \frac{\sum_{i=1}^n w_i \mathbb{I}\{c_b(x_i) \neq y_i\}}{\sum_{i=1}^n w_i}$ // error of c_b
[Update]: $\alpha_b = \log(\frac{1-\epsilon_b}{\epsilon_b})$ // always > 0, final weight of c_b
[Reweight]: $w_i \leftarrow w_i \exp(\alpha_b \mathbb{I}\{c_b(x_i) \neq y_i\})$ normalize!
Return $\hat{c}_{\text{ADA}}(x) = \text{sign}(\sum_{b=1}^B \alpha_b c_b(x))$

7 Deep Learning

Sigmoid: $\sigma(x) = \frac{1}{1+\exp(-x)} = \frac{e^x}{e^x+1}$
$\sigma'(x) = \sigma(x)(1-\sigma(x)) = \sigma(x)\sigma(-x)$
Softmax: $y_i \propto \exp(\beta z_i)$
Backpropagation: Gradient: $\frac{\partial \ell}{\partial w_{jk}} = \delta_j^{(l)} v_k^{(l-1)}$

7.1 Variational Autoencoders

Def: $\xrightarrow[\text{prior}]{p_{\theta'}(z)} \mathcal{Z} \xrightarrow[\text{likelihood}]{\text{dec}_\theta(z) = p_\theta(x|z)} \mathcal{X} \xrightarrow[\text{approx. posterior}]{\text{enc}_\phi(x) = q_\phi(z|x)} \mathcal{Z}$
sample/obs. $\mathcal{X}_i$ from latent representation $\mathcal{Z}$
Train: $\max_{\theta',\theta,\phi} \sum_i \underbrace{\log p_{\theta',\theta}(x_i)}_{(*)\text{ indep. of } Z}$

$$(*) = \mathbb{E}_{Z\sim q_\phi(\cdot|x_i)}\left[\log\left(\frac{p_{\theta',\theta}(x_i,Z)}{p_{\theta',\theta}(Z|x_i)} \frac{q_\phi(Z|x_i)}{q_\phi(Z|x_i)}\right)\right]$$
$$= \overbrace{\mathbb{E}[\log p_\theta(x_i \mid Z)] - D_{\text{KL}}\big(q_\phi(\cdot \mid x_i) \,\|\, p_{\theta'}(\cdot)\big)}^{\mathcal{L}(x_i,\theta,\phi) \equiv \textbf{ELBO} = \text{Infomax} - \text{Regularisation term}}$$
$$+ D_{\text{KL}}\big(q_\phi(\cdot \mid x_i) \,\|\, p_{\theta',\theta}(\cdot \mid x_i)\big) \ge \mathcal{L}(x_i,\theta,\phi)$$

Train: $\theta^*, \phi^* = \arg\max_{\theta,\phi} \mathcal{L}(x_i,\theta,\phi)$

Requirements for good representation:
- **informative:** $\theta^* = \arg\max_\theta I(X;Z)$
  $= \arg\max_\theta \mathbb{E}_{X,Z}[\log p(X \mid Z)] - const_{\text{w.r.t. }\theta}$
  $\approx \arg\max_\theta \sum_i \mathbb{E}_{Z|X}[\log p(x_i \mid Z)]$
- **disentangled:** components in $\mathcal{Z}$ associated with distinct feature in $\mathcal{X}$ (see D_{KL} in ELBO).
- **robust:** noise in $\mathcal{Z}$ doesn't substantially affect $\mathcal{X}$ (and vice versa). → choice of approx. post.!

7.2 Transformers

Attention: Query: $Q = EW^Q$, Key: $K = EW^K$, Value: $V = EW^V$, embedding E.
$A = \text{softmax}(\frac{QK^\top}{\sqrt{d}})V$, d: E-dim.
- **Self-Attention:** Q, K, V from same sequence
- **Cross-Att.:** Q from target, K, V from source.
- **Multi-headed:** Multiple weights, Stack A_i's.
- **Masked Self-Attention:** Future tokens masked ($-\infty$ in upper triangle of $P = QK^\top$)

Pos. Encoding: $p_{i,2j} = \sin(\alpha^{-2j} i)$, $p_{i,2j+1} = \cos(\alpha^{-2j} i)$, for $j \le d/2$, $\alpha = 10^{4/d}$.

7.3 Stable Diffusion

Forward Process: For $t \le T$, $\beta_t > 0$, let $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{s\le t} \alpha_s$
$q(x_t|x_{t-1}) = \mathcal{N}(x_t | \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I)$
$q(x_t|x_0) = \mathcal{N}(x_t | \sqrt{\bar{\alpha}_t}\, x_0, (1-\bar{\alpha}_t) I)$
Reverse Process: UNet predicts noise $\epsilon_\theta(x_t, t)$ added to form x_t from x_0. Then reconstruct x_{t-1}:
$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\sqrt{1-\alpha_t}}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t,t)\right) + \sigma_t z$, $z \sim \mathcal{N}(0, I)$

8 Clustering

***k*-means:** $\arg\min_\theta \sum_{i\le n} \|\boldsymbol{x}_i - \theta_{c(\boldsymbol{x}_i)}\|^2$

8.1 Mixture Models

Assume: $\boldsymbol{x} \sim p(\boldsymbol{x} \mid \pi_{1\dots k}, \theta_{1\dots k}) = \sum_{c\le k} \pi_c p(\boldsymbol{x} \mid \boldsymbol{\theta}_c)$
Find: $\hat{\theta} = \arg\max_\theta p(\mathcal{X} \mid \pi, \theta) = \prod_x p(x \mid \pi, \theta)$
Gaussian Mixtures: → $p(\boldsymbol{x} \mid \theta_c) = p(\boldsymbol{x} \mid \boldsymbol{\mu}, \Sigma)$
Introduce *latent indicator variables* for mode assignments $M_{xc} \in \{0,1\}$. Then, the **log-likelihood:**
$L(\mathcal{X}, \boldsymbol{M} \mid \boldsymbol{\theta}) = \sum_{\boldsymbol{x}} \sum_{c\le k} M_{xc} \log(\pi_c p(\boldsymbol{x} \mid \boldsymbol{\theta}_c))$

8.1.1 EM-Algorithm for Gaussian Mixtures

E-step: Calculate
$Q(\boldsymbol{\theta}; \boldsymbol{\theta}^{(t)}) = \mathbb{E}_{\boldsymbol{M}|\mathcal{X},\boldsymbol{\theta}^{(t)}}[L(\mathcal{X}, \boldsymbol{M} \mid \boldsymbol{\theta})] = \dots$
$= \sum_{\boldsymbol{x}} \sum_{c\le k} \big(\underbrace{\mathbb{E}_{\boldsymbol{M}|\mathcal{X},\boldsymbol{\theta}^{(t)}}[M_{xc}]} \cdot \log \pi_c p(\boldsymbol{x} \mid \boldsymbol{\theta}_c)\big)$
where $\gamma_{xc} = \frac{p(\boldsymbol{x}|c,\boldsymbol{\theta}^{(t)})\, p(c|\boldsymbol{\theta}^{(t)})}{p(\boldsymbol{x}|\boldsymbol{\theta}^{(t)})}$, $\sum_{c\le k} \gamma_{xc} = 1$
M-step: $\boldsymbol{\theta}^{(t+1)} \in \arg\max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta}; \boldsymbol{\theta}^{(t)})$
s.t. $\sum_c \pi_c = 1$. Solve via Lagrangian, yields
$\pi_c = \frac{1}{|\mathcal{X}|} \sum_{\boldsymbol{x}} \gamma_{xc}$, $\boldsymbol{\mu}_c = \frac{\sum_x \gamma_{xc} \boldsymbol{x}}{\sum_x \gamma_{xc}}$, $\sigma_c^2 = \frac{\sum_x \gamma_{xc} (\boldsymbol{x} - \boldsymbol{\mu}_c)^2}{\sum_x \gamma_{xc}}$

8.2 Non-parametric Bayesian Methods

$\text{Dir}(\boldsymbol{x} \mid \boldsymbol{\alpha}) = \frac{1}{B(\boldsymbol{\alpha})} \prod_{k=1}^n x_k^{\alpha_k - 1}$, $B(\boldsymbol{\alpha}) = \frac{\prod_{k=1}^n \Gamma(\alpha_k)}{\Gamma(\sum_{k=1}^n \alpha_k)}$
Rewrite **Finite mixture models:**
$p(x) = \sum_{k=1}^K \pi_k p(x \mid \theta_k) = \int p(x \mid \theta) G(\theta)\, \mathrm{d}\theta$
where $G(\theta) = \sum_{k=1}^\infty \pi_k \delta_{\theta_k}(\theta) \leftarrow$ discrete distr.

Stick-breaking process:
Draw $\theta_k \sim H$ and $\beta_k \sim \text{Beta}(1, \alpha)$ for $k = 1, 2, \dots$
$\pi_k = \beta_k (1 - \sum_{i=1}^{k-1} \pi_i) \implies \pi = \{\pi_k\}_{k=1}^\infty \sim \text{GEM}(\alpha)$
$\implies \sum_{k=1}^\infty \pi_k \delta_{\theta_k}(\theta) = G(\theta) \sim \text{DP}(\alpha, H)$
Sample $\theta^{(1)}, \theta^{(2)}, \dots$ from G. Denote $\theta^{(i)} = \theta_{k_i}$.
$\implies \theta^{(i)}, \theta^{(j)}$ with $k_i = k_j$ belong to same "cluster"
Chinese Rest. Process:
$P(\text{cust}_{n+1} \text{ joins table } \tau \mid \mathcal{P}) = \begin{cases} \frac{|\tau|}{\alpha+n} & \text{if } \tau \in \mathcal{P}, \\ \frac{\alpha}{\alpha+n} & \text{new table} \end{cases}$
$P(\text{partition } \mathcal{P}) = \frac{\alpha^{|\mathcal{P}|}}{\alpha^{(n)}} \prod_{\tau \in \mathcal{P}} (|\tau| - 1)!$
expec. #clusters: $\mathbb{E}[1] = \sum_{i\le N} \frac{\alpha}{\alpha+i} \sim \mathcal{O}(\alpha \log N)$
De Finetti: $(X_1, \dots, X_n)$ are inftly **exchangeable** RVs *if* $P(X_1, \dots, X_n) = \int \big(\prod_{i=1}^n p(X_i \mid G)\big) \mathrm{d}P(G)$

8.2.1 Finite GMM

1. Cluster centers: $\mu_k \sim \mathcal{N}(\mu_0, \sigma_0)$
2. Prob's of clusters: $\pi_{1\dots K} \sim \text{Dir}(\alpha_{1\dots K})$
3. Cluster assignments: $z_i \sim \text{Categorical}(\pi_{1\dots K})$
4. Coordinates of data: $x_i \sim \mathcal{N}(\mu_{z_i}, \sigma_{z_i})$

8.2.2 DP Mixture Model (DP-GMM)

1. Cluster centers: $\mu_k \sim \mathcal{N}(\mu_0, \sigma_0)$, $k = 1, 2, \dots$
2. Prob's of clusters: $\pi = (\pi_1, \pi_2, \dots) \sim \text{GEM}(\alpha)$
3. Cluster assignments: $z_i \sim \text{Categorical}(\pi)$
4. Coordinates of data: $x_i \sim \mathcal{N}(\mu_{z_i}, \sigma)$, $i = 1 \dots N$

Fitting a DP-MM: Collapsed Gibbs sampler
$p(z_i = k \mid \boldsymbol{z}_{-i}, \boldsymbol{x}, \alpha, \boldsymbol{\mu}) \propto \text{Prior} \times \text{Likelihood}$
$\propto \begin{cases} \frac{|\boldsymbol{x}_{-i,k}|}{\alpha+N-1} p(x_i | \boldsymbol{x}_{-i,k}, \boldsymbol{\mu}) & \text{for existing } k \\ \frac{\alpha}{\alpha+N-1} p(x_i | \boldsymbol{\mu}) & \text{otw.} \end{cases}$
$\boldsymbol{x}_{-i,c} := \{x_j \mid z_j = c, j \neq i\}$ data assigned to clust. c

9 PAC Learning

Want: Distribution indep. error guarantees!
Expec./**Gener. error:** $\mathcal{R}(\hat{c}_n) = \boldsymbol{P}_{X,Y}(\hat{c}_n(x) \neq c(x))$
Empirical error: $\hat{\mathcal{R}}_n(\hat{c}_n) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{\hat{c}_n(x_i) \neq y_i\}}$
PAC learnable: $\mathcal{A}$ can learn a concept class $\mathcal{C}$ from $\mathcal{H}$ if, given a suff. large sample, it outputs a hypothesis that generalizes well w/ high prob.
(1) $0 < \epsilon < 1/2$, $0 < \delta < 1/2$, (2) $\boldsymbol{P}_{X,Y}$ on $\mathcal{X} \times \{0,1\}$:
If $n \ge poly(1/\epsilon, 1/\delta, dim(\mathcal{X}))$,
Then $\boldsymbol{P}_{X,Y}\left(\mathcal{R}(\hat{c}_n) - \inf_{c\in\mathcal{C}} \mathcal{R}(c) \le \epsilon\right) \ge 1 - \delta$.
If $\mathcal{A}$ runs in time polynomial in $1/\epsilon$ and $1/\delta$, we say that $\mathcal{C}$ is **efficiently PAC learnable**.

9.1 VC Inequality $\boldsymbol{P}(\dots \ge \epsilon) \le \dots \le \delta$

Select ERM: $\hat{c}_n^* = \arg\min_{c\in\mathcal{C}} \hat{\mathcal{R}}_n(c)$
Under uniform convergence:
$\boldsymbol{P}\left(\mathcal{R}(\hat{c}_n^*) - \inf_{c\in\mathcal{C}} \mathcal{R}(c) > \epsilon\right) \le \boldsymbol{P}\left(\sup_{c\in\mathcal{C}} \left|\hat{\mathcal{R}}_n(c) - \mathcal{R}(c)\right| > \frac{\epsilon}{2}\right)$
- $|\mathcal{C}|$ **Finite:** $\boldsymbol{P}(\sup|\cdots| > \epsilon) \le 2|\mathcal{C}| \exp(-2n\epsilon^2)$
- $|\mathcal{C}|$ **Unbounded:** $\boldsymbol{P}(\cdots) \le 9n^{\text{VC}_\mathcal{C}} \exp(-\frac{n\epsilon^2}{32})$

9.2 Rectangle Learning

$\boldsymbol{P}((\hat{c}_n^* > \epsilon) \le |\mathcal{C}| \cdot (1-\epsilon)^n \le |\mathcal{C}| \cdot \exp(-n\epsilon) < \delta$
Union bound: $\boldsymbol{P}(\bigcup_i T_i) \le \sum_i \boldsymbol{P}(T_i)$

10 Appendix

Complete the square: If $p(\boldsymbol{x}) \propto \exp(-\frac{1}{2}\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} + \boldsymbol{x}^\top \boldsymbol{b})$, then $p(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{x} \mid \boldsymbol{A}^{-1}\boldsymbol{b}, \boldsymbol{A}^{-1})$
Constrained optimisation:
primal: $\min_{\boldsymbol{x}} f(\boldsymbol{x})$ s.t. $g_i(\boldsymbol{x}) = 0$; $h_j(\boldsymbol{x}) \le 0$
Lagrangian: with each $\alpha_j \ge 0$
$\mathcal{L}(\boldsymbol{x}, \lambda, \alpha) = f(\boldsymbol{x}) + \sum_i \lambda_i g_i(\boldsymbol{x}) + \sum_j \alpha_j h_j(\boldsymbol{x})$
Solve: $\frac{\partial \mathcal{L}}{\partial \boldsymbol{x}} = 0$; $g_i(\boldsymbol{x}) = 0$; $\alpha_j \ge 0$; $h_j(\boldsymbol{x}) \le 0$
If **Slater's cond.** holds, $\exists \boldsymbol{x} : g_i(\boldsymbol{x}) = 0, h_j(\boldsymbol{x}) < 0$, then we can solve the *dual* instead:
$\max_{\lambda,\alpha}\{\min_{\boldsymbol{x}} \mathcal{L}(\boldsymbol{x}, \lambda, \alpha)\}$ s.t. $\alpha_j \ge 0$
Solve: $\frac{\partial \mathcal{L}}{\partial \boldsymbol{x}} = 0$; $\frac{\partial \mathcal{L}}{\partial \lambda} = 0$; $\alpha_j h_j(\boldsymbol{x}) = 0$; $\alpha_j \ge 0$
Metrics: $acc = \frac{TP+TN}{n}$ $prec = \frac{TP}{TP+FP}$ $FPR = \frac{FP}{FP+TN}$
$Recall/TPR = \frac{TP}{TP+FN}$ $balanced\ acc = \frac{1}{n}\sum_i TPR_i$
$F1{-}score = \frac{2TP}{2TP+FP+FN}$ $ROC = FPR/TPR$
Conditional Gaussians:
$P_{X,Y} = \begin{bmatrix} \boldsymbol{X} \\ \boldsymbol{Y} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \boldsymbol{\mu}_X \\ \boldsymbol{\mu}_Y \end{bmatrix}, \begin{bmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{bmatrix}\right)$, Σ_{ij} p.s.d.
$\implies Y|X \sim \mathcal{N}(\tilde{\mu}, \tilde{\Sigma})$, where $\tilde{\boldsymbol{\mu}} = \boldsymbol{\mu}_Y + \Sigma_{YX}\Sigma_{XX}^{-1}(X - \boldsymbol{\mu}_X)$, $\tilde{\Sigma} = \Sigma_{YY} - \Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XY}$